# Unicritical compressed shifts contain the required pseudohyperbolic disk through degree twelve

Candidate partial result for arXiv:2112.06321

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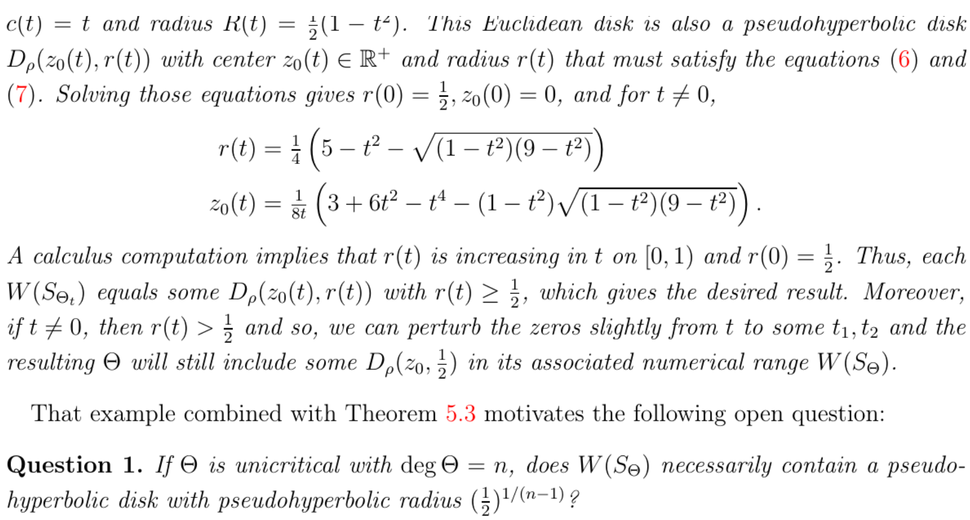
**Status.** Candidate partial result, likely valid. The source question is answered for all unicritical degrees $2 \le n \le 12$. The new part of this packet is the uniform-in-the-zero proof for $6 \le n \le 12$. No claim is made for $n \ge 13$.

## 1 Source question and scope

Bickel and Gorkin ask the following in Question 1 of Section 6, PDF page 28 of [1].

> If $\Theta$ is unicritical with $\deg \Theta = n$, does $W(S_\Theta)$ necessarily contain a pseudohyperbolic disk with pseudohyperbolic radius $(1/2)^{1/(n-1)}$?



*$c(t) = t$ and radius $R(t) = \frac{1}{2}(1-t^2)$. This Euclidean disk is also a pseudohyperbolic disk $D_\rho(z_0(t), r(t))$ with center $z_0(t) \in \mathbb{R}^+$ and radius $r(t)$ that must satisfy the equations (6) and (7). Solving those equations gives $r(0) = \frac{1}{2}$, $z_0(0) = 0$, and for $t \neq 0$,*

$$r(t) = \tfrac{1}{4}\left(5 - t^2 - \sqrt{(1-t^2)(9-t^2)}\right)$$
$$z_0(t) = \tfrac{1}{8t}\left(3 + 6t^2 - t^4 - (1-t^2)\sqrt{(1-t^2)(9-t^2)}\right).$$

*A calculus computation implies that $r(t)$ is increasing in $t$ on $[0,1)$ and $r(0) = \frac{1}{2}$. Thus, each $W(S_{\Theta_t})$ equals some $D_\rho(z_0(t), r(t))$ with $r(t) \ge \frac{1}{2}$, which gives the desired result. Moreover, if $t \neq 0$, then $r(t) > \frac{1}{2}$ and so, we can perturb the zeros slightly from $t$ to some $t_1, t_2$ and the resulting $\Theta$ will still include some $D_\rho(z_0, \frac{1}{2})$ in its associated numerical range $W(S_\Theta)$.*

That example combined with Theorem 5.3 motivates the following open question:

**Question 1.** *If $\Theta$ is unicritical with $\deg \Theta = n$, does $W(S_\Theta)$ necessarily contain a pseudohyperbolic disk with pseudohyperbolic radius $(\frac{1}{2})^{1/(n-1)}$?*

Figure 1: The complete source question and its immediate context, rendered from PDF page 28 of [1].

The source proves the requested assertion through degree four. Bickel, Corbett, Glenning, Guan, and Vollmayr-Lee later proved the degree-five case, in fact with radius $\sqrt{3}/2$, in Theorem 2.1 of [2]. Their Remark following Corollary 2.2 reports numerical evidence in degrees $6, 7, 8$, and $11$, but not a proof. The result below rigorously covers every degree from six through twelve.

## 2 Idea of the proof

For a unicritical product with repeated zero $t \in [0,1)$, the compressed shift is represented by a KMS matrix

$$M_{n,t} = tI + (1-t^2)A_{n,t}, \qquad (A_{n,t})_{ij} = \begin{cases} (-t)^{j-i-1}, & i < j, \\ 0, & i \ge j. \end{cases}$$

Bickel and Gorkin give an explicit trigonometric curve $f_{n,t}(s)$ inside $W(A_{n,t})$. Its values at $s=0$ and $s=\pi$ have midpoint $b_{n,t}$ and half-separation $L_{n,t} > 0$.

The key new point is an exact lower bound

$$|f_{n,t}(s) - b_{n,t}| \ge \kappa_n L_{n,t} \quad (0 \le t \le 1,\ s \in \mathbb{R}),$$

for explicit rational $\kappa_n < 1$. Since the centered curve has winding number one, this puts the whole disk $D(b_{n,t}, \kappa_n L_{n,t})$ in its convex hull. Affine rescaling puts a corresponding Euclidean disk in $W(M_{n,t})$. A second exact inequality shows that this Euclidean disk has pseudohyperbolic radius at least $2^{-1/(n-1)}$.

Both inequalities reduce to polynomial positivity on a unit box. The included checker proves positivity using exact rational Bernstein coefficients, with exact isolating intervals and error bounds for the two algebraic constants. Thus the computation is a finite proof certificate, not sampled numerical evidence.

## 3 The partial theorem

Recall that a degree-$n$ unicritical Blaschke product has the form

$$\Theta(z) = \lambda \left( \frac{z - z_0}{1 - \overline{z_0} z} \right)^n, \qquad |\lambda| = 1, \quad z_0 \in \mathbb{D}.$$

**Theorem 1** (Degrees six through twelve)**.** *Let $6 \le n \le 12$, and let $\Theta$ be a degree-$n$ unicritical finite Blaschke product. Then $W(S_\Theta)$ contains a pseudohyperbolic disk of radius*

$$q_n = 2^{-1/(n-1)}.$$

*Consequently, together with [1, 2], the source Question 1 has an affirmative answer for every degree $2 \le n \le 12$.*

## 4 The KMS curve

Multiplying $\Theta$ by a unimodular constant does not change the model operator. The rotation identity for the numerical range in [1] reduces the repeated zero to $t = |z_0| \in [0,1)$. In the Takenaka–Malmquist–Walsh basis, the matrix of $S_\Theta$ is $M_{n,t}$ as displayed above.

Put $\vartheta_n = \pi/(n+1)$ and

$$a_{n,k} = \frac{(n-k)\cos(k\vartheta_n)\sin\vartheta_n + \sin((n-k)\vartheta_n)}{(n+1)\sin\vartheta_n} \qquad (1 \le k \le n-1).$$

Proposition 5.1 of [1] gives the curve

$$f_{n,t}(s) = \sum_{k=1}^{n-1} a_{n,k}(-t)^{k-1} e^{iks} \in W(A_{n,t}). \tag{1}$$

For completeness, this follows by testing $A_{n,t}$ on the unit vector with coordinates

$$x_\ell(s) = \sqrt{\frac{2}{n+1}}\sin(\ell\vartheta_n)e^{i(\ell-1)s}.$$

Define

$$b_{n,t} = \sum_{\substack{1\le k\le n-1\\ k \text{ even}}} a_{n,k}(-t)^{k-1}, \tag{2}$$

$$L_{n,t} = \sum_{\substack{1\le k\le n-1\\ k \text{ odd}}} a_{n,k}t^{k-1}. \tag{3}$$

All $a_{n,k}$ are positive, so $L_{n,t} > 0$. Directly from (1),

$$f_{n,t}(0) = b_{n,t} + L_{n,t}, \qquad f_{n,t}(\pi) = b_{n,t} - L_{n,t}.$$

**Lemma 2** (Certified curve separation)**.** *For every $t \in [0,1]$ and $s \in \mathbb{R}$,*

$$|f_{n,t}(s) - b_{n,t}| \ge \kappa_n L_{n,t},$$

*where the constants are as follows.*

| $n$ | $\kappa_n$ | $n$ | $\kappa_n$ |
|---|---|---|---|
| *6* | 124/125 | *10* | 499/500 |
| *7* | 997/1000 | *11* | 999/1000 |
| *8* | 249/250 | *12* | 999/1000 |
| *9* | 499/500 | | |

*The inequality is strict after squaring and moving the right side to the left.*

*Proof.* Set $c_n = \cos\vartheta_n$ and $x = \cos s$. The identity

$$\frac{\sin((j+1)\vartheta_n)}{\sin\vartheta_n} = U_j(c_n)$$

expresses every $a_{n,k}$ as a polynomial in $c_n$ with rational coefficients. Because all coefficients of $f_{n,t}$ are real,

$$P_n(t,x,c_n) = |f_{n,t}(s) - b_{n,t}|^2 - \kappa_n^2 L_{n,t}^2$$

is a polynomial in $t$, $x$, and $c_n$. Substitute $x = 2y - 1$.

The script `code/verify_degrees_6_to_12.py` constructs this polynomial exactly. It isolates $c_n$ by its exact minimal polynomial in a rational interval of width $10^{-32}$. At the rational midpoint it subdivides $[0,1]^2$ into rational boxes. On every box all tensor-product Bernstein coefficients of the transformed polynomial are nonnegative. The polynomial is therefore nonnegative there, since it is a convex combination of those coefficients in the Bernstein basis. Finally, an exact coefficient $\ell^1$ bound for $\partial P_n/\partial c$ dominates the whole isolating-interval error. The resulting lower bounds remain strictly positive. Table 1 records the certificate sizes and decimal displays of the exact rational lower bounds. ∎

**Lemma 3** (Disk from the winding curve)**.** *For $6 \le n \le 12$ and $0 \le t < 1$,*

$$D(b_{n,t}, \kappa_n L_{n,t}) \subseteq W(A_{n,t}).$$

Table 1: Exact certificate diagnostics. The displayed lower bounds are decimal approximations of positive rational numbers computed by the checker.

| $n$ | curve boxes | max depth | curve lower bound | radius lower bound |
|---|---|---|---|---|
| 6 | 7 | 6 | $1.2935 \cdot 10^{-2}$ | $1.8253 \cdot 10^{-3}$ |
| 7 | 5 | 4 | $5.1136 \cdot 10^{-3}$ | $1.4739 \cdot 10^{-3}$ |
| 8 | 5 | 4 | $4.4936 \cdot 10^{-4}$ | $2.8709 \cdot 10^{-3}$ |
| 9 | 5 | 4 | $3.6144 \cdot 10^{-3}$ | $1.7864 \cdot 10^{-3}$ |
| 10 | 7 | 6 | $3.6788 \cdot 10^{-3}$ | $3.5949 \cdot 10^{-3}$ |
| 11 | 5 | 4 | $1.8651 \cdot 10^{-3}$ | $3.1819 \cdot 10^{-3}$ |
| 12 | 9 | 6 | $1.8845 \cdot 10^{-3}$ | $2.7708 \cdot 10^{-3}$ |

*Proof.* The curve (1) is closed and lies in the convex set $W(A_{n,t})$. By Lemma 2, its translate $f_{n,t}(s) - b_{n,t}$ never vanishes. Its winding number about zero is constant under the homotopy $t \in [0,1]$. At $t = 0$ the curve is $a_{n,1}e^{is}$, so the winding number is one.

If $|w - b_{n,t}| < \kappa_n L_{n,t}$, Lemma 2 allows $w$ to be moved to $b_{n,t}$ without crossing the curve. Hence the curve has winding number one about $w$. A point outside the convex hull of a closed curve can be strictly separated from it by a line, and the curve then has winding number zero about that point. Thus every such $w$ lies in the convex hull of the curve, which is contained in $W(A_{n,t})$. Closure gives the claimed closed disk. □

## 5 Pseudohyperbolic radius and proof of the theorem

By Lemma 3 and the affine relation between $M_{n,t}$ and $A_{n,t}$,

$$D(C_{n,t}, R_{n,t}) \subseteq W(M_{n,t}), \tag{4}$$

where

$$C_{n,t} = t + (1 - t^2)b_{n,t}, \qquad R_{n,t} = (1 - t^2)\kappa_n L_{n,t}. \tag{5}$$

Every Euclidean disk contained in $\mathbb{D}$ is a pseudohyperbolic disk. If its pseudohyperbolic radius is $r \in (0,1)$, then

$$r + \frac{1}{r} = \frac{R^2 - |C|^2 + 1}{R}; \tag{6}$$

see Equation (13) of [2].

**Lemma 4** (Certified radius comparison)**.** *For $6 \le n \le 12$, $0 \le t < 1$, and $q_n = 2^{-1/(n-1)}$, the quantities in (5) satisfy*

$$q_n\big(R_{n,t}^2 - C_{n,t}^2 + 1\big) - (q_n^2 + 1)R_{n,t} \le 0. \tag{7}$$

*Proof.* The left side of (7) factors exactly as

$$(t^2 - 1)Q_n(t, c_n, q_n).$$

The verifier isolates both algebraic constants by their exact minimal polynomials. At the rational interval midpoints, every univariate Bernstein coefficient of $Q_n$ on $[0,1]$ is positive, without subdivision. Exact coefficient $\ell^1$ bounds for the derivatives with respect to $c_n$ and $q_n$ dominate the isolating-interval errors. The final exact rational lower bounds for $Q_n$ are the last column of Table 1. Hence $Q_n > 0$, while $t^2 - 1 < 0$ for $t < 1$, which proves (7). □

*Proof of Theorem 1.* Let $r_{n,t}$ be the pseudohyperbolic radius of the disk in (4). Dividing (7) by the positive quantity $q_n R_{n,t}$ gives

$$\frac{R_{n,t}^2 - C_{n,t}^2 + 1}{R_{n,t}} \leq q_n + \frac{1}{q_n}.$$

The function $u \mapsto u + u^{-1}$ is strictly decreasing on $(0, 1]$. Equation (6) therefore implies $r_{n,t} \geq q_n$. This proves the theorem for a repeated zero $t \in [0, 1)$. The rotation reduction at the start of Section 4 handles every $z_0 \in \mathbb{D}$. □

# 6 Verification, novelty check, and limitations

The proof certificate is reproducible with

```
conda run --no-capture-output -n sandbox python \
  code/verify_degrees_6_to_12.py
```

It prints one `PASS` line for each $n = 6, \ldots, 12$. The complete recorded output and an explanation of why the checks are exact appear in `verification_report.txt`. The smaller `code/verify_degree6.py` independently specializes the construction to the first previously unresolved degree.

A bounded primary-source search was performed on 11–12 August 2026. It included arXiv searches for the exact phrases "pseudohyperbolic disk criterion", "level set Crouzeix", "unicritical Blaschke numerical range", and the radius $\cos(\pi/(n+1))$, together with citation tracing from arXiv:2112.06321. The search found arXiv:2312.04537, which explicitly proves degree five and reports only numerical evidence in several higher degrees. No primary arXiv source proving degrees six through twelve or all degrees was located. This is a bounded search, not a definitive priority claim.

The limitation is essential: the certificate is finite and degree-specific. The computations strongly suggest a uniform inequality, but no proof of that uniform inequality was obtained. In particular, this packet does not answer the source question for $n \geq 13$.

**Human-review recommendation.** The result is suitable for expert review as a candidate partial theorem. The two most important manual checks are Lemma 3 and the direction of the monotonicity argument following Equation (6). The exact verifier should then be rerun in a clean environment.

# References


[1] K. Bickel and P. Gorkin, *Blaschke Products, Level Sets, and Crouzeix's Conjecture*, arXiv:2112.06321 (2021).

[2] K. Bickel, G. Corbett, A. Glenning, C. Guan, and M. Vollmayr-Lee, *Crouzeix's conjecture, compressions of shifts, and classes of nilpotent matrices*, arXiv:2312.04537 (2023).